Computer Architecture: Orientation, Introduction, and Computer 101A

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Perspectives

- Who are you?
- What are you here for?
- What is your most important thing?

This class

- Computer and how it works.
- Focus on the CPU.
- Main topics
  - An entire system: processor, memory, i/o.
  - Processor: instruction set
  - Processor: computer arithmetic
  - Processor: datapath and control
  - Processor: pipeline
- You are a part of the team, our team.

Our team

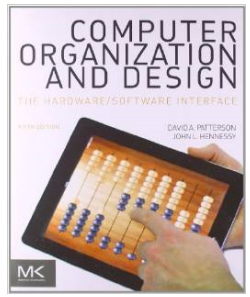
- Everyone is our team member.
- Every member is crucial.
- Our goal is to train you all to be good engineers.
- Our team will work together to achieve the goal.

Evaluation

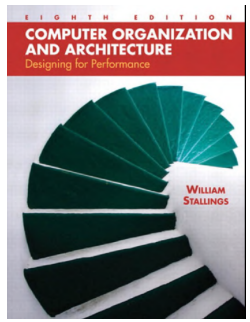
- Work 50%
  - Character/Tenacity
  - Meeting/Participation
  - Workshop/Commitment
- Exam 50%

Main materials

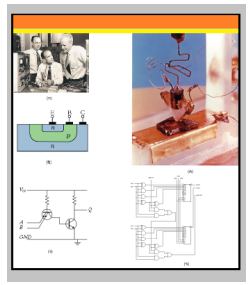
David A. Patterson and John L. Hennessy, Computer Organization and Design: The Hardware/Software Interface. The Morgan Kaufmann, October 10, 2013



William Stallings, Computer Organization and Architecture. Pearson (ฉบับแปล สัสยุทธ์ สว่างวรรณ 2003)

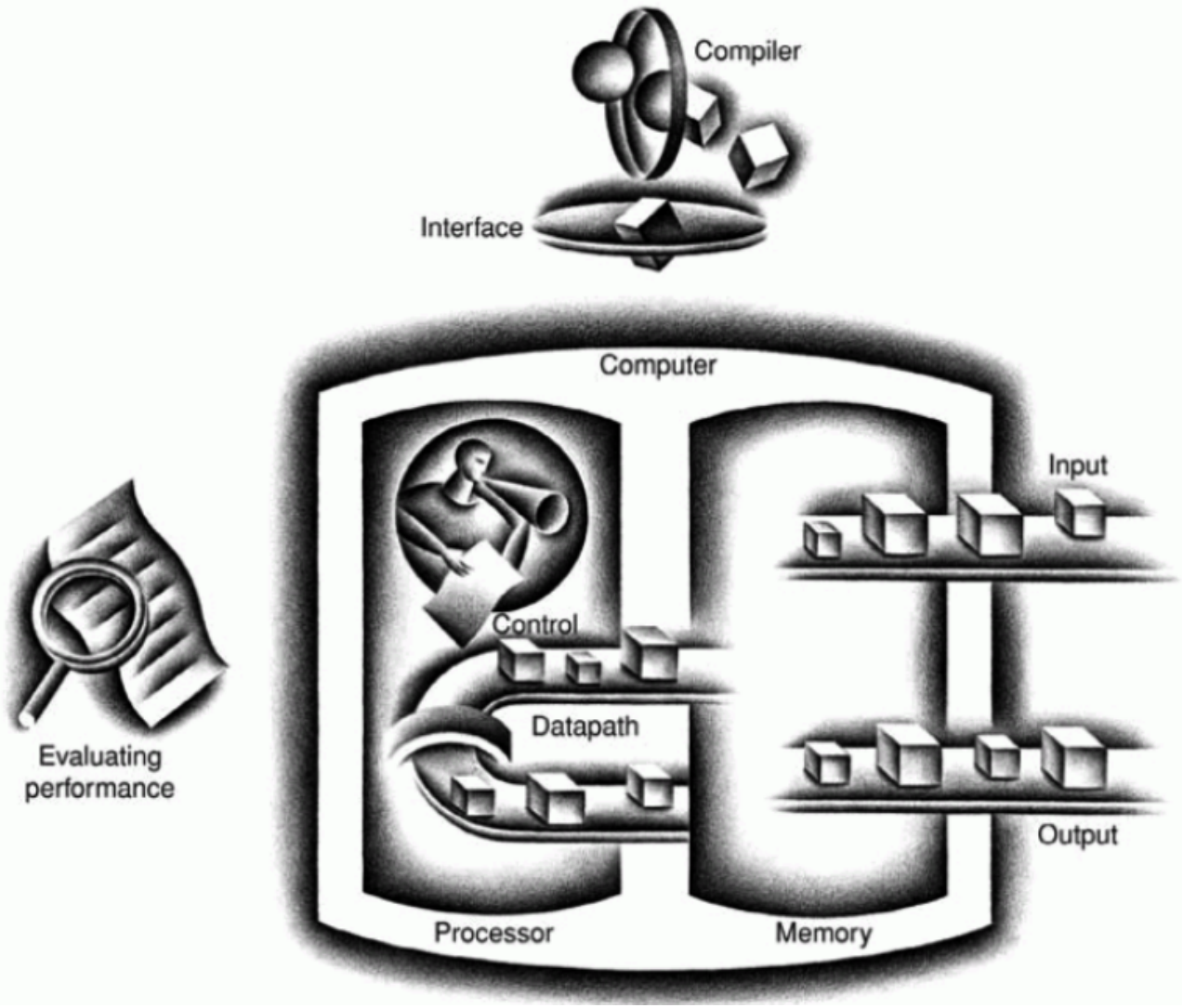


เอกสารประกอบคำสอน สถาปัตยกรรมคอมพิวเตอร์. https://sites.google.com/site/kkucoeclasses/coa

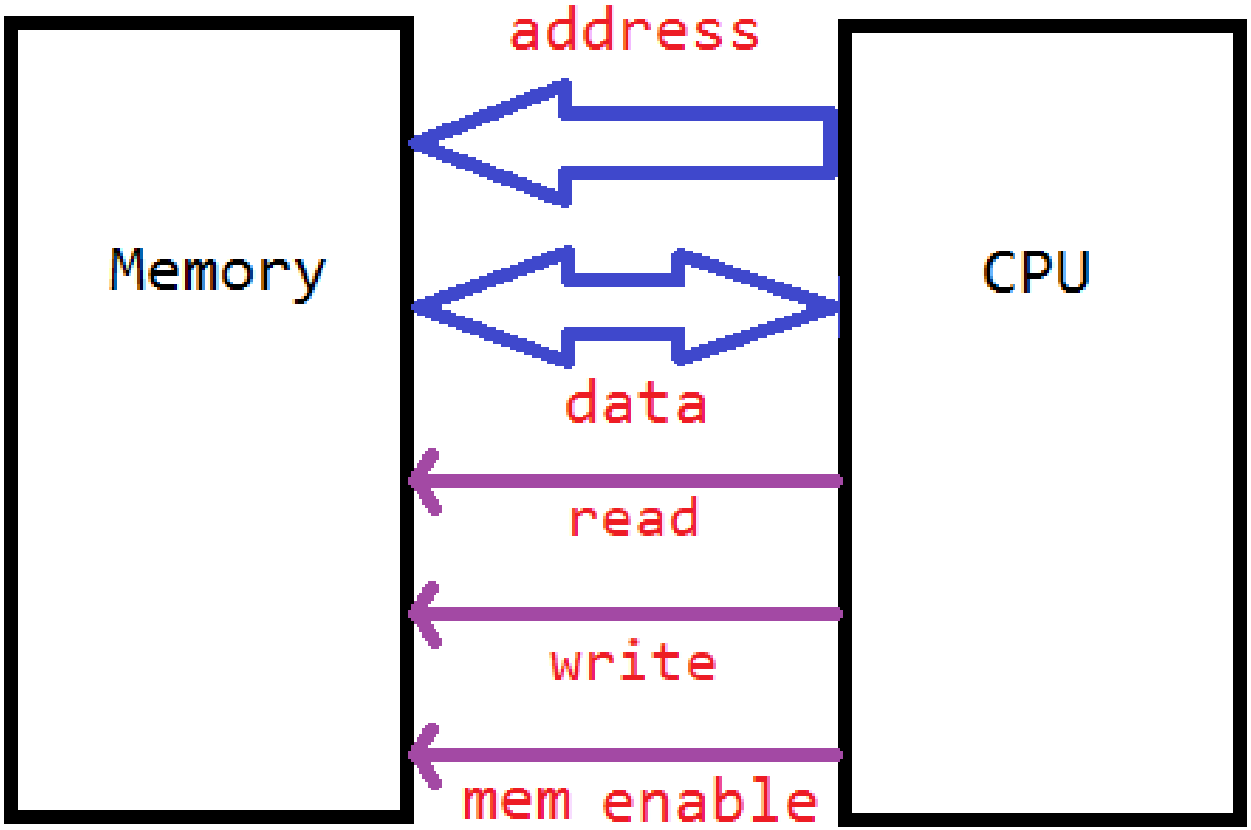


Key Components of a Computer

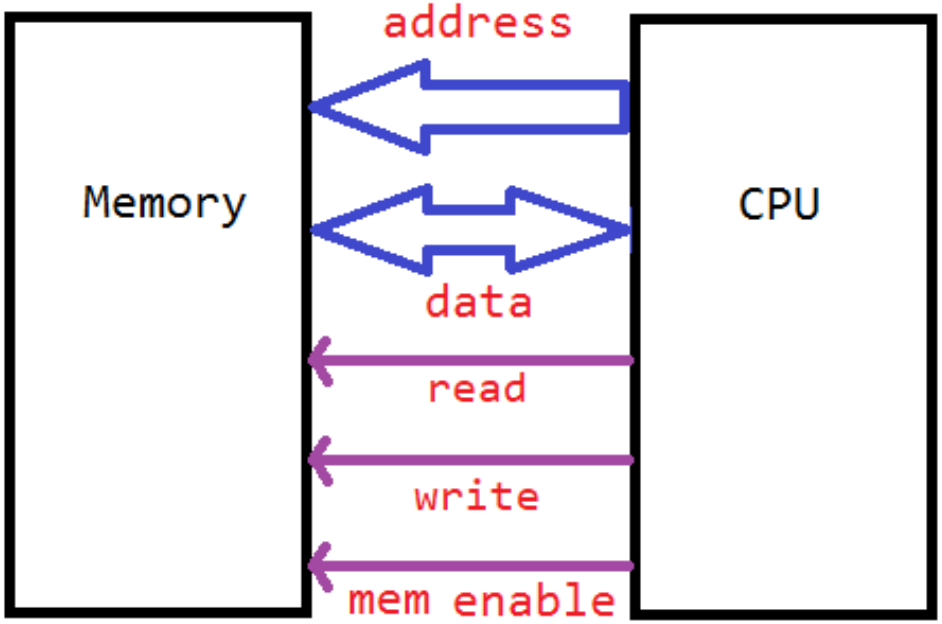
- Processor
  - Datapath
  - Control
- Memory
- I/O

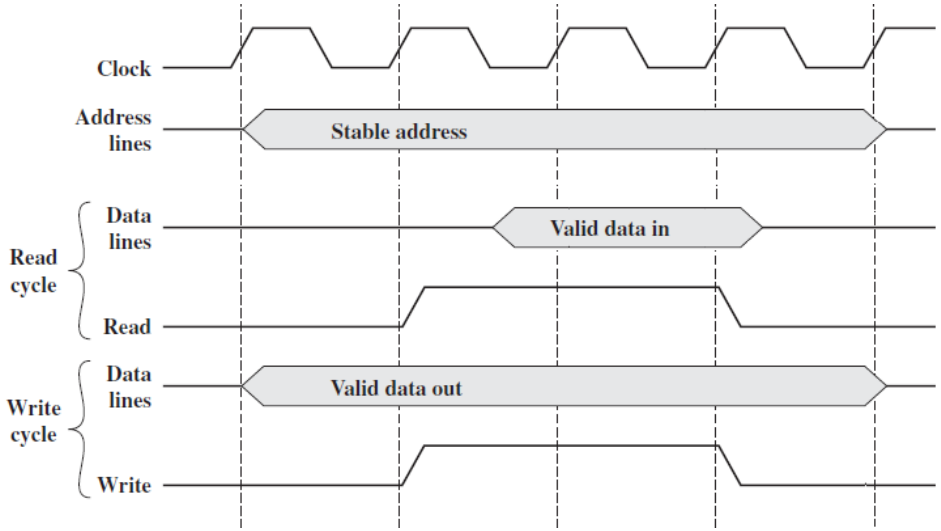


Memory and processor (1)



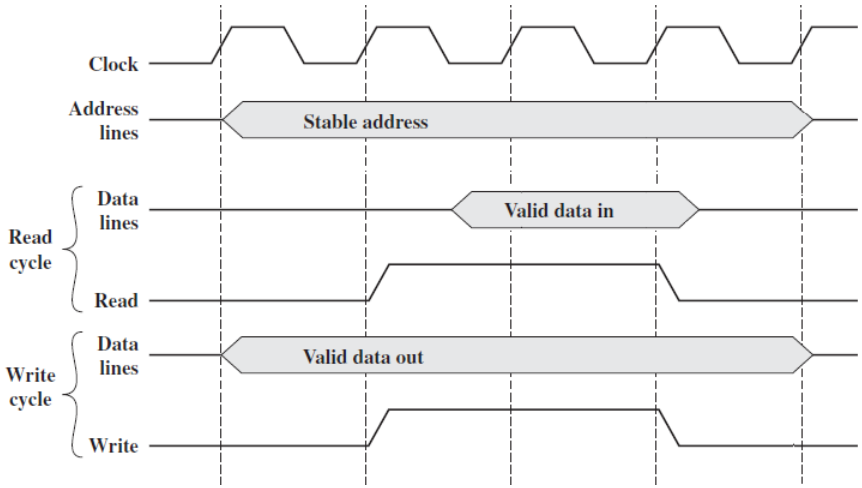
Memory and processor (2)



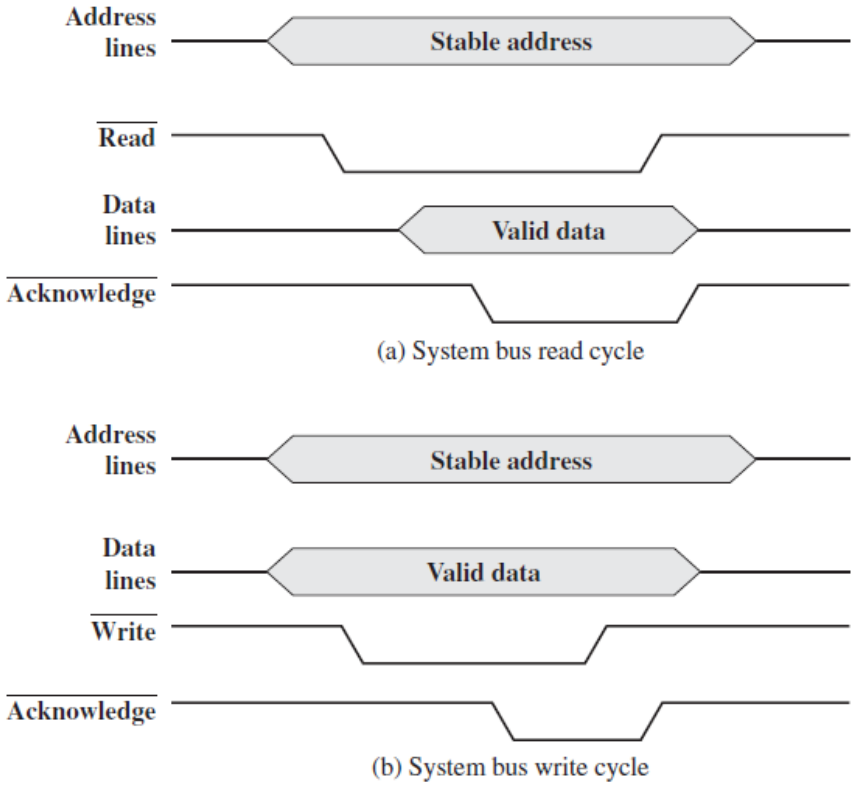


Synchronous v.s. Asynchronous

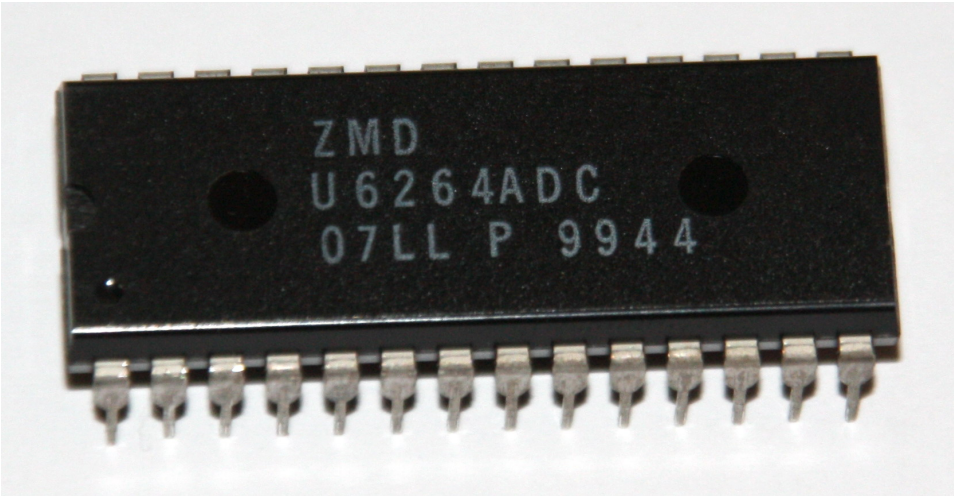
Synchronous Operation



Asynchronous Operation



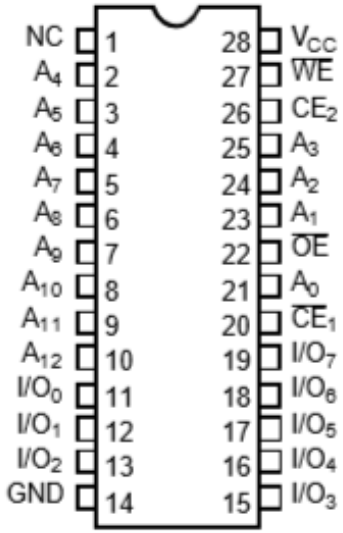
Example: CY6264, 8K x 8 Static RAM



src: https://commons.wikimedia.org/wiki/File:ZMD_U6264_SRAM_1999.jpg

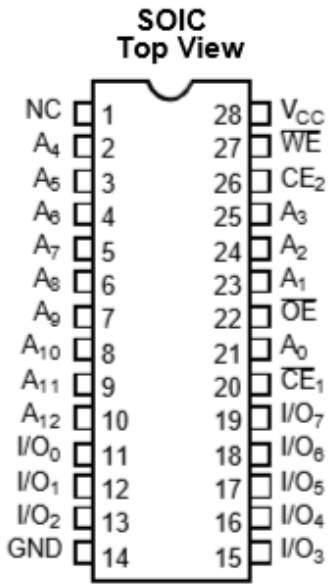
Pin Configuration

SOIC
Top View



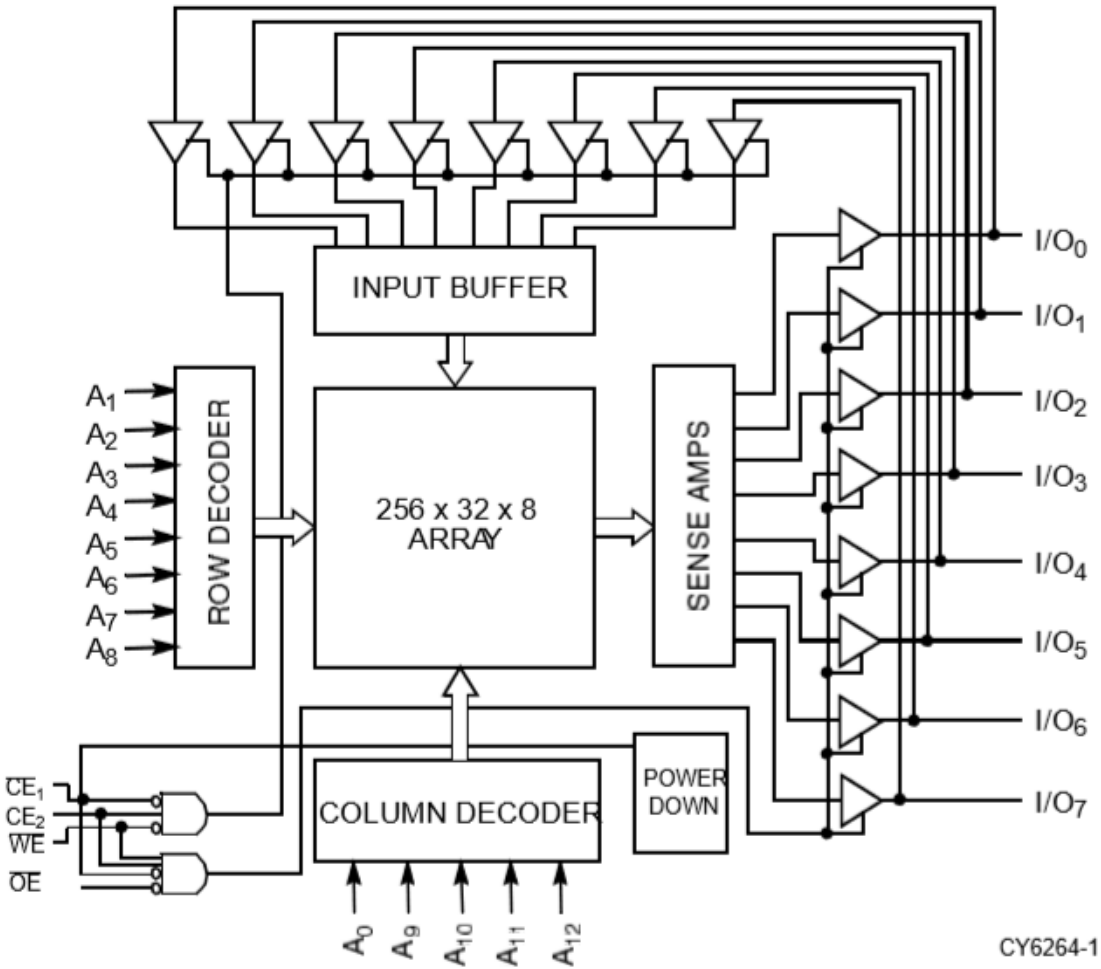
CY6264, 8K x 8 Static RAM

Pin Configuration



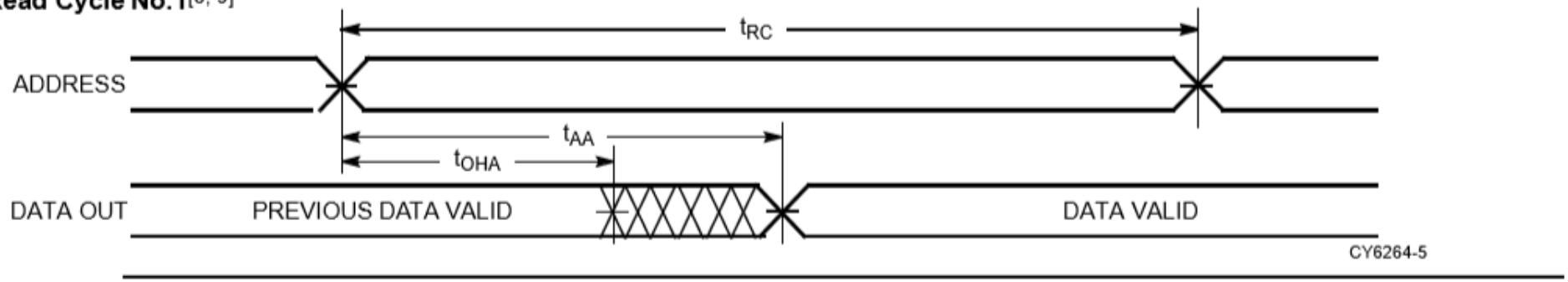
src: CY6264 data sheet

Logic Block Diagram

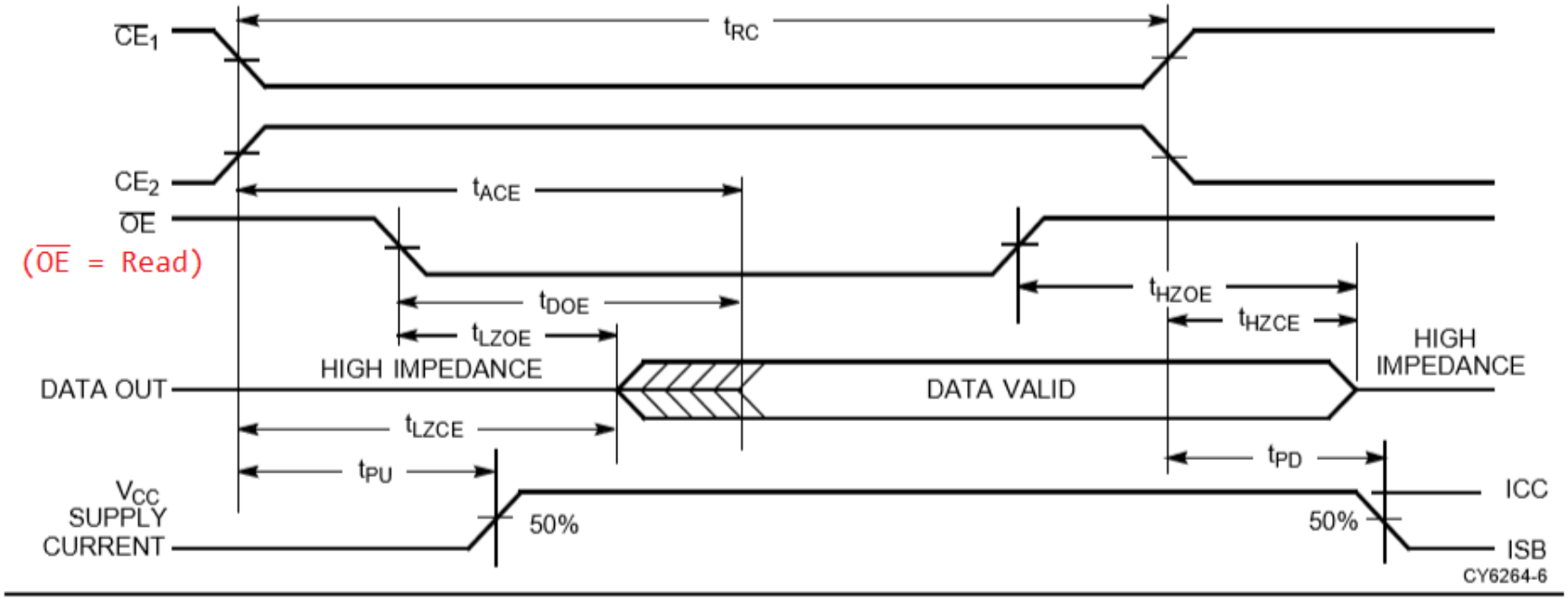


CY6264: read cycle

Read Cycle No.1[8, 9]



Read Cycle No. 2[10, 11]



(Adjusted horizonal alignment between 2 sub-diagrams for easy reading)

CY6264: switching characteristics: read

Switching Characteristics Over the Operating Range[4]

Parameter	Description	6264-55		6264-70		Unit
		Min.	Max.	Min.	Max.	
READ CYCLE						
t_{RC}	Read Cycle Time	55		70		ns
t_{AA}	Address to Data Valid		55		70	ns
t_{OHA}	Data Hold from Address Change	5		5		ns
t_{ACE1}	$\overline{CE}_1$ LOW to Data Valid		55		70	ns
t_{ACE2}	CE_2 HIGH to Data Valid		40		70	ns
t_{DOE}	$\overline{OE}$ LOW to Data Valid		25		35	ns
t_{LZOE}	$\overline{OE}$ LOW to Low Z	3		5		ns
t_{HZOE}	$\overline{OE}$ HIGH to High Z[5]		20		30	ns
t_{LZCE1}	$\overline{CE}_1$ LOW to Low Z[6]	5		5		ns
t_{LZCE2}	CE_2 HIGH to Low Z	3		5		ns
t_{HZCE}	$\overline{CE}_1$ HIGH to High Z[5, 6] CE_2 LOW to High Z		20		30	ns
t_{PU}	$\overline{CE}_1$ LOW to Power-Up	0		0		ns
t_{PD}	$\overline{CE}_1$ HIGH to Power-Down		25		30	ns

Read-cycle questions

Given 6264-55,

- How fast can this memory be read?

  Ans:

- What is the shortest duration between chip enable (CE and $\bar{CE}$) and read ($\bar{OE}$) signals?

  Ans:

- How long has the address to be sent before the read signal?

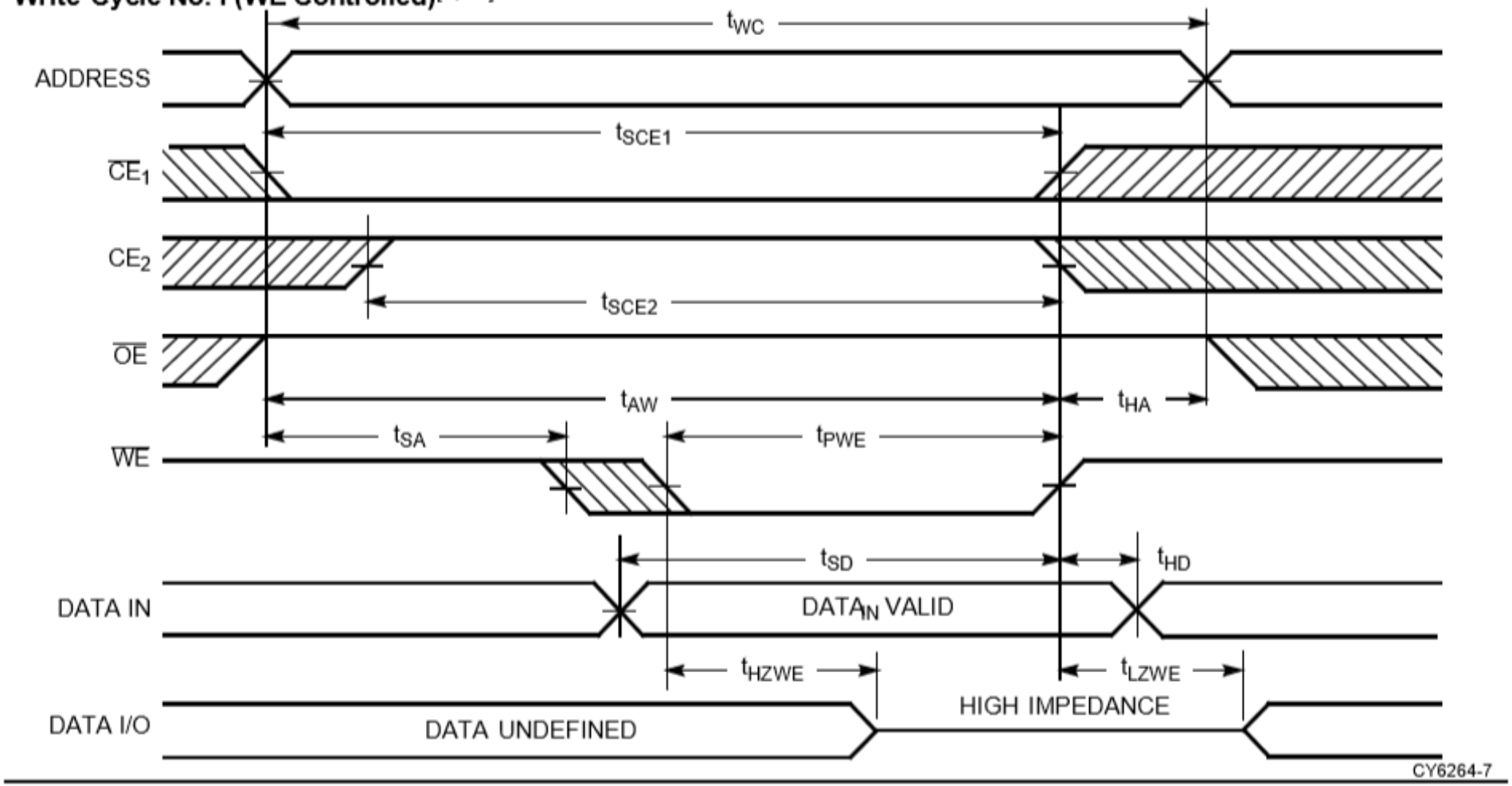
  Ans:

- After read signal is sent, how long can we expect to see the data?

  Ans:

CY6264: write cycle

Write Cycle No. 1 ($\overline{WE}$ Controlled)[9, 11]



Notes:

8. Device is continuously selected. $\overline{OE}$, $\overline{CE}$ = V_{IL}. CE_2 = V_{IH}.
9. Address valid prior to or coincident with $\overline{CE}$ transition LOW.
10. $\overline{WE}$ is HIGH for read cycle.
11. Data I/O is High Z if $\overline{OE}$ = V_{IH}, $\overline{CE_1}$ = V_{IH}, or $\overline{WE}$ = V_{IL}.

CY6264: switching characteristics: write

Switching Characteristics Over the Operating Range[4]

Parameter	Description	6264-55		6264-70		Unit
		Min.	Max.	Min.	Max.	
WRITE CYCLE[7]						
t_{WC}	Write Cycle Time	50		70		ns
t_{SCE1}	$\overline{CE}_1$ LOW to Write End	40		60		ns
t_{SCE2}	CE_2 HIGH to Write End	30		50		ns
t_{AW}	Address Set-Up to Write End	40		55		ns
t_{HA}	Address Hold from Write End	0		0		ns
t_{SA}	Address Set-Up to Write Start	0		0		ns
t_{PWE}	$\overline{WE}$ Pulse Width	25		40		ns
t_{SD}	Data Set-Up to Write End	25		35		ns
t_{HD}	Data Hold from Write End	0		0		ns
t_{HZWE}	$\overline{WE}$ LOW to High Z[5]		20		30	ns
t_{LZWE}	$\overline{WE}$ HIGH to Low Z	5		5		ns

Write-cycle questions

Given 6264-55,

- How fast can this memory be written?

  Ans:
- How long has the write signal to be held?

  Ans:
- How long has the address to be sent before the write signal ends?

  Ans:
- How long has the data to be sent before the write signal ends?

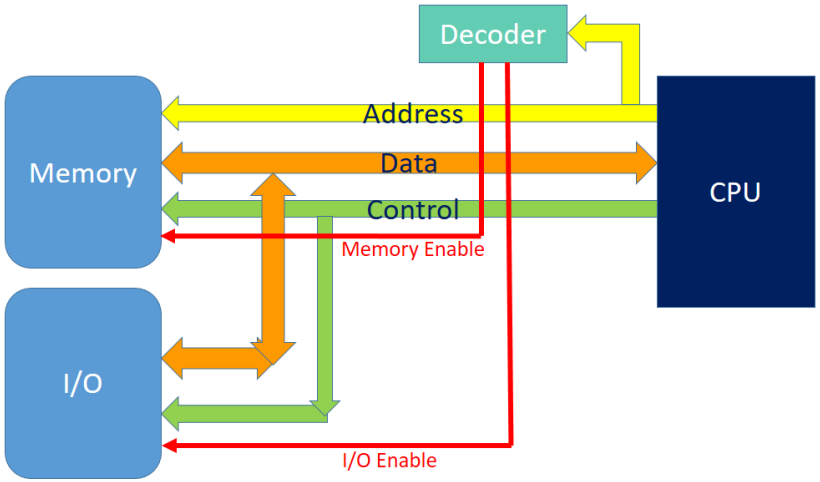
  Ans:
- How long has the address and data to be held after the write signal ends?

  Ans:

Accessing I/O

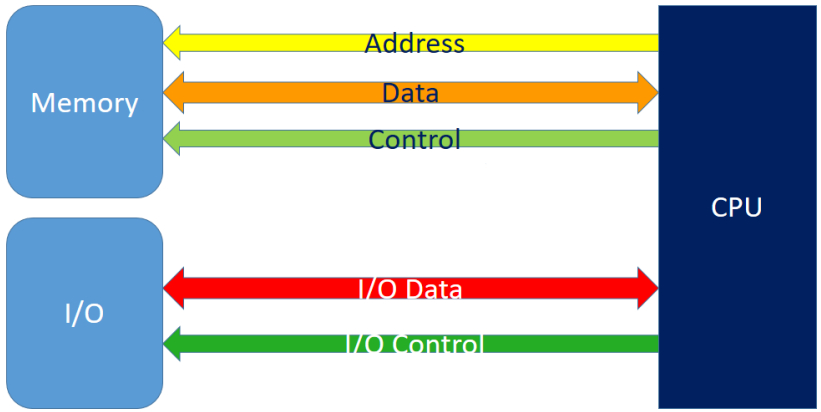
Memory mapped I/O

- Single address space for memory location and I/O devices. Status and data registers of I/O modules treated as memory locations.
- Same machine instructions to access both memory and I/O



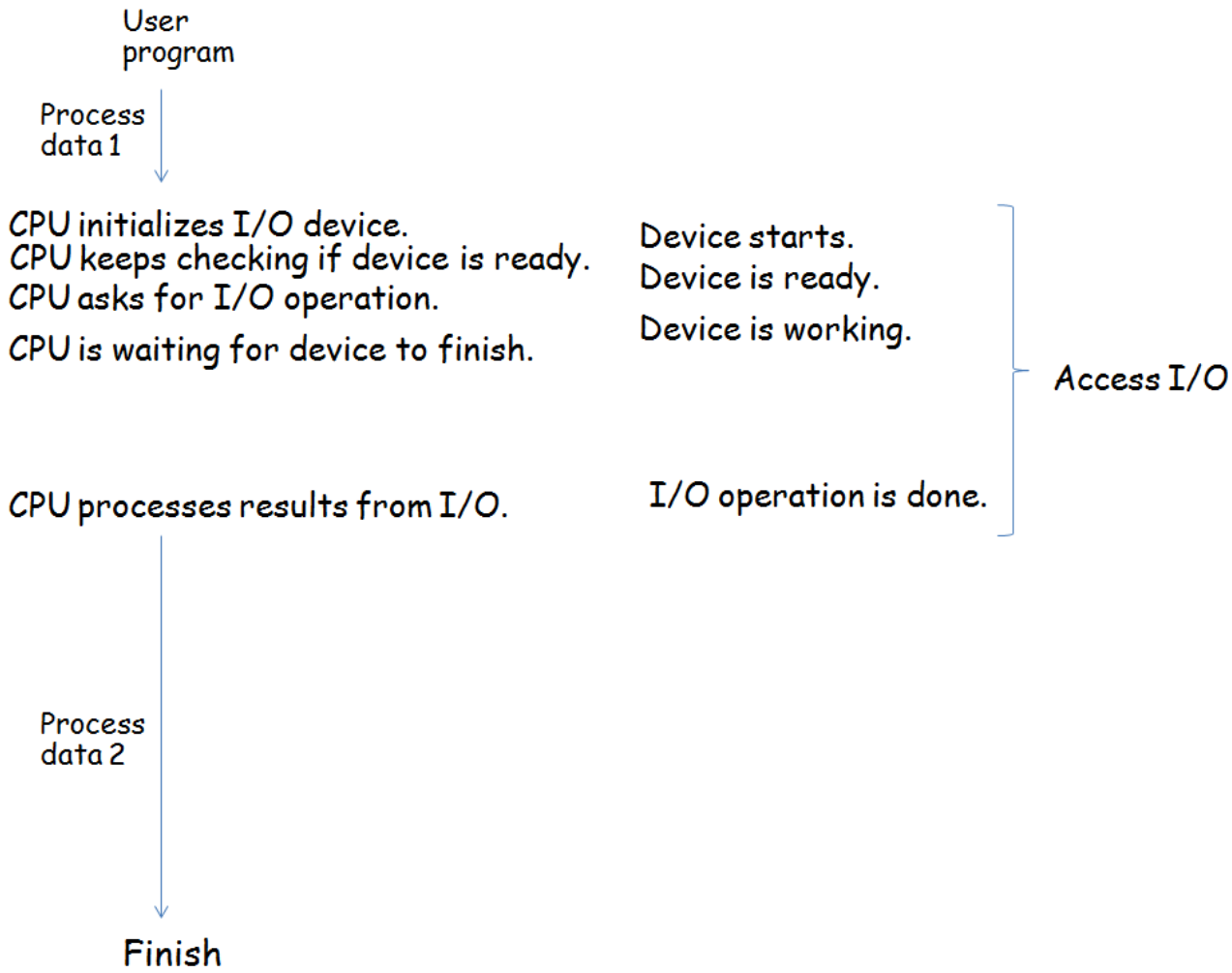
Isolated I/O or dedicated I/O

- Memory read and write lines + input and output lines
- dedicated instructions for I/O

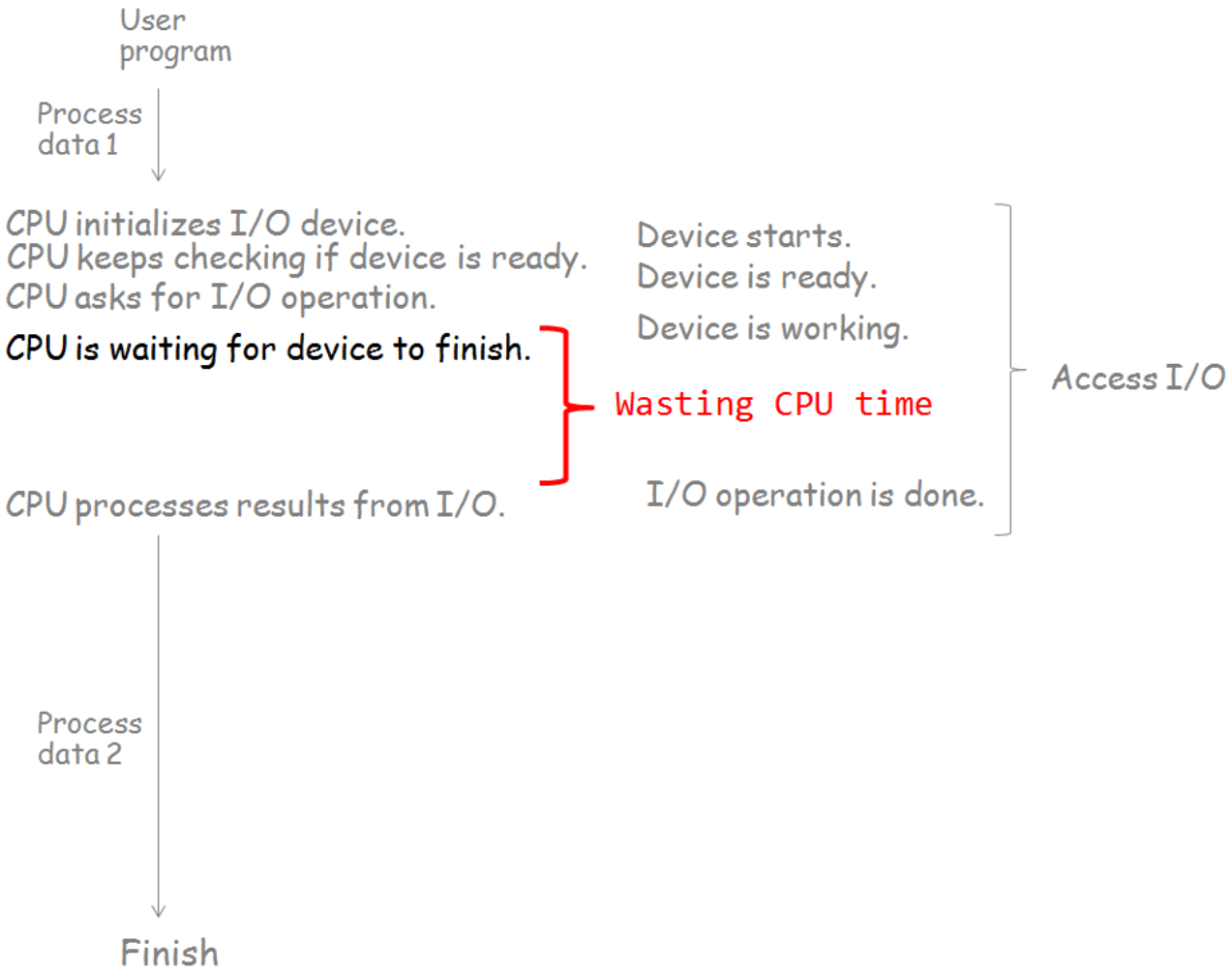


Speed mismatch

- I/O is much slower than CPU $\Rightarrow$ Loop to check if it is done. (Polling)

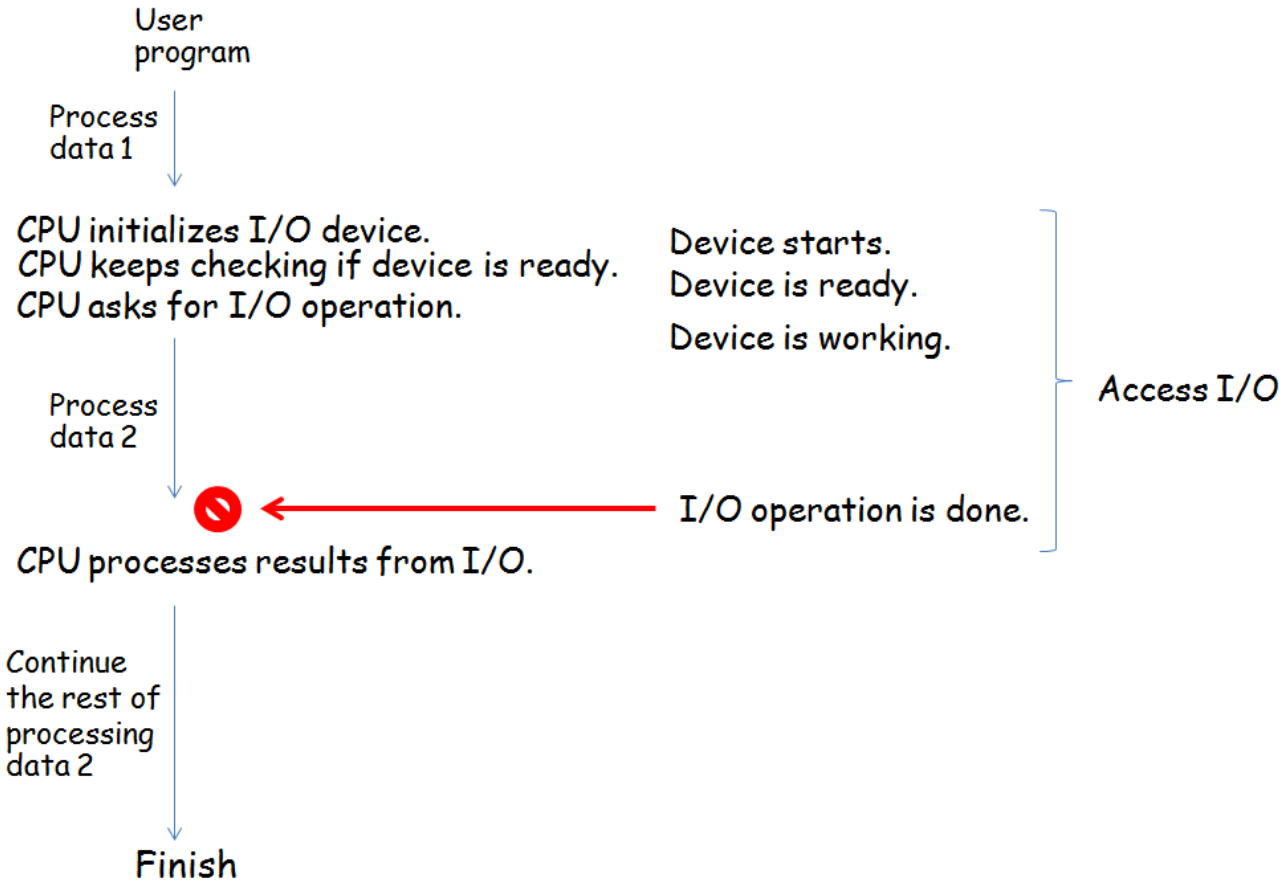


Polling I/O is inefficient



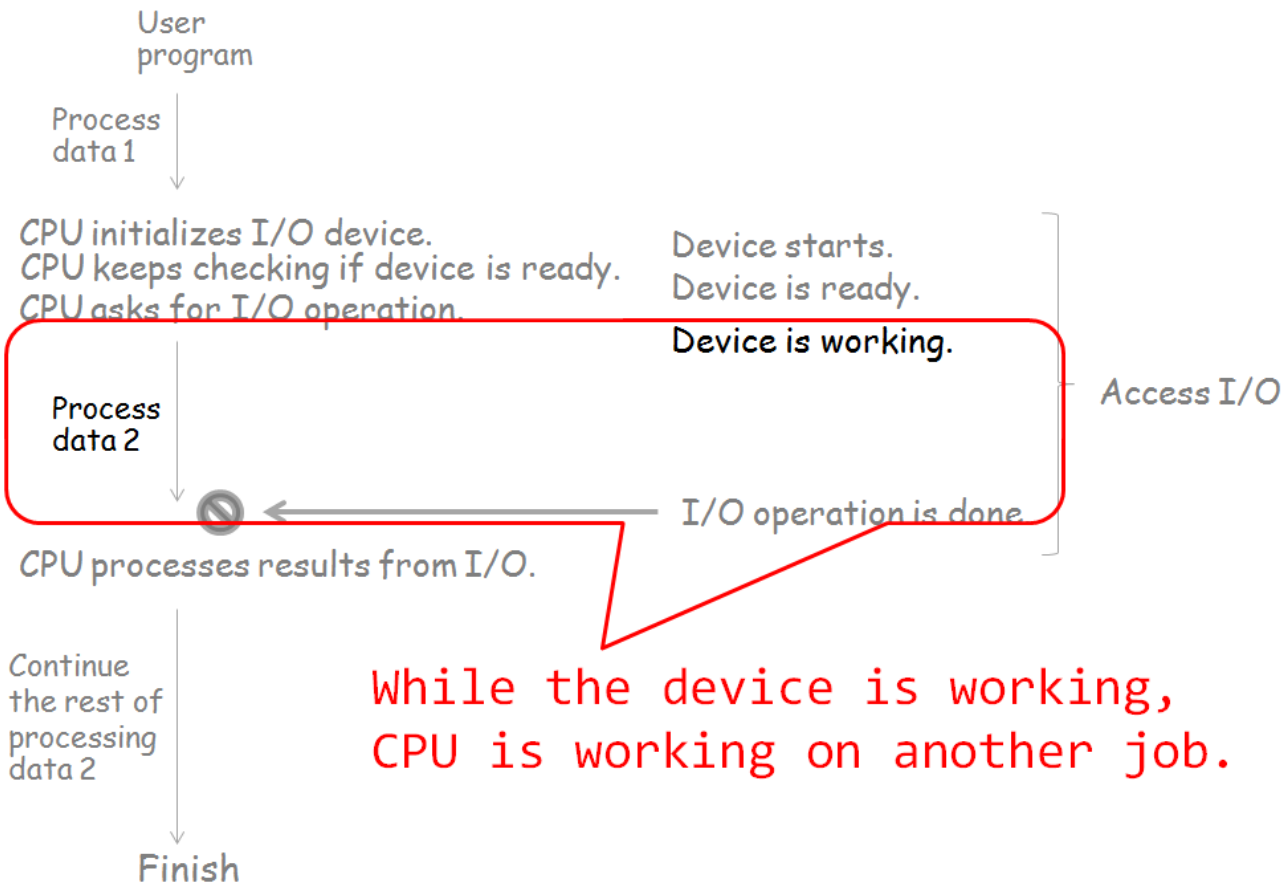
Interrupt (1)

- Interrupt: a mechanism to inform CPU for attention



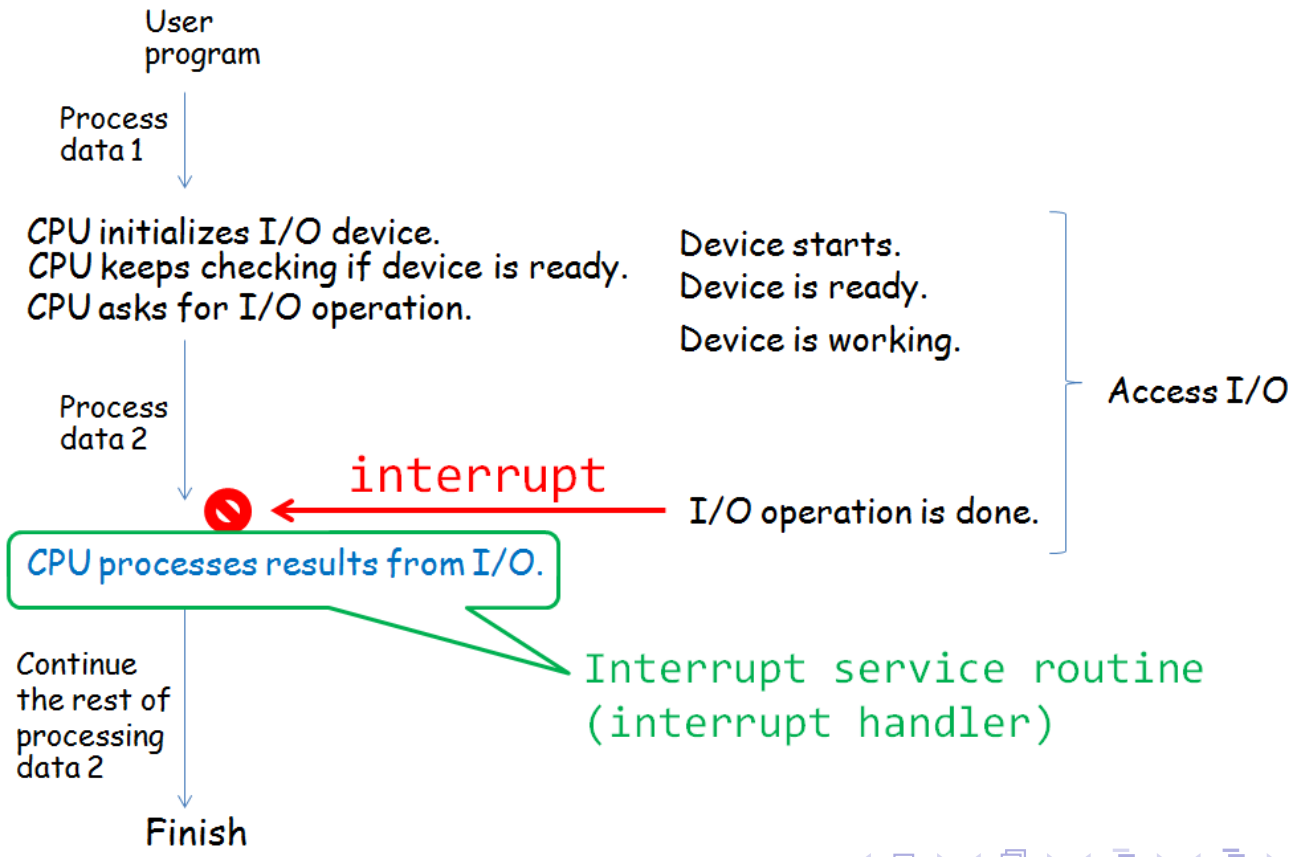
Interrupt (2)

- Interrupt allows CPU to do something else while waiting



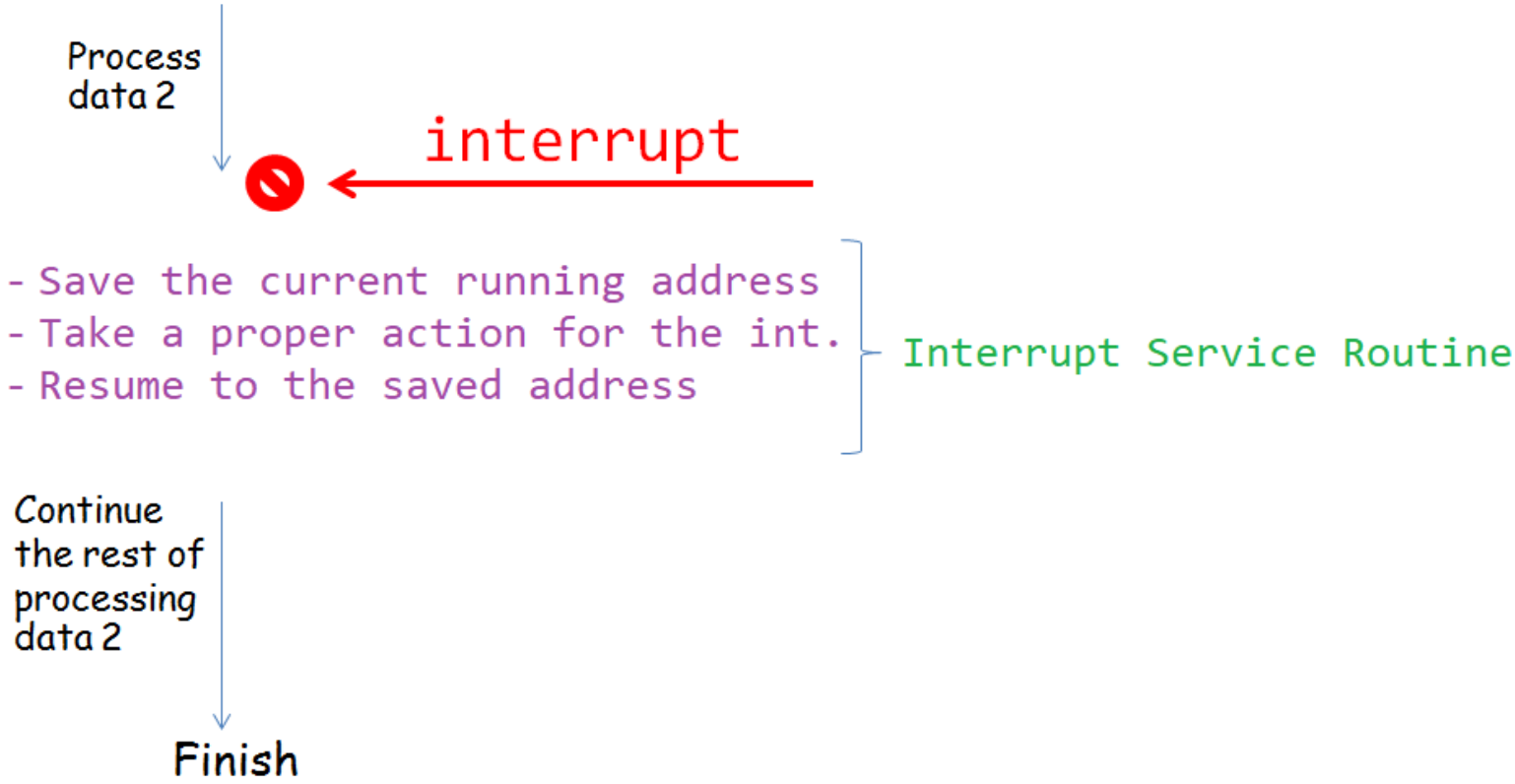
Interrupt handler (1)

- When CPU is interrupted, it stops what it is doing and run an interrupt service routine. Interrupt service routine is a program (software) and must be prepared beforehand.



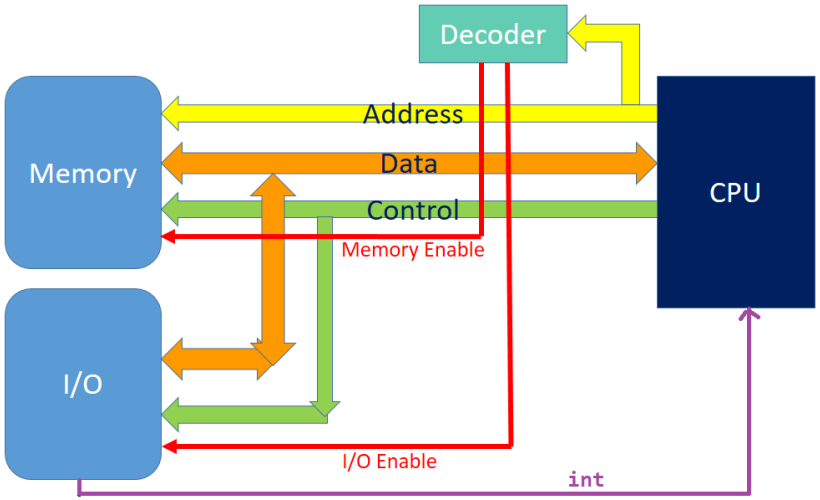
Interrupt handler (2)

• Interrupt service routine must specify a proper action for the interrupt. Once the interrupt service routine is finished, CPU must resume to do whatever it left off for the interrupt.

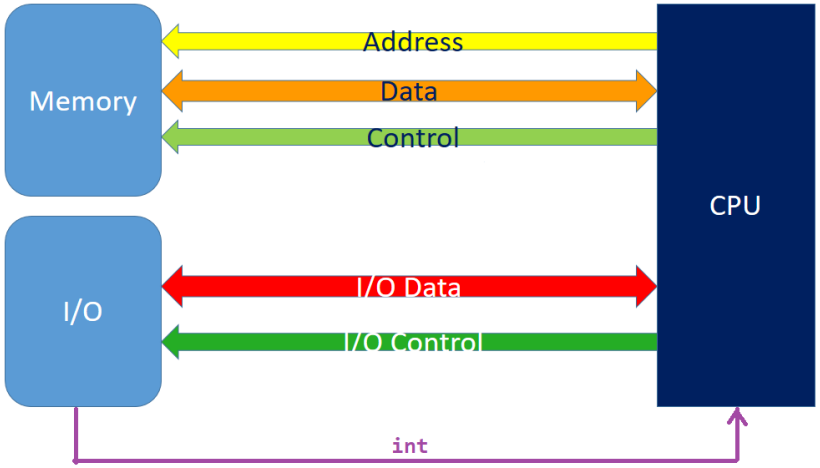


CPU and connections with interrupt

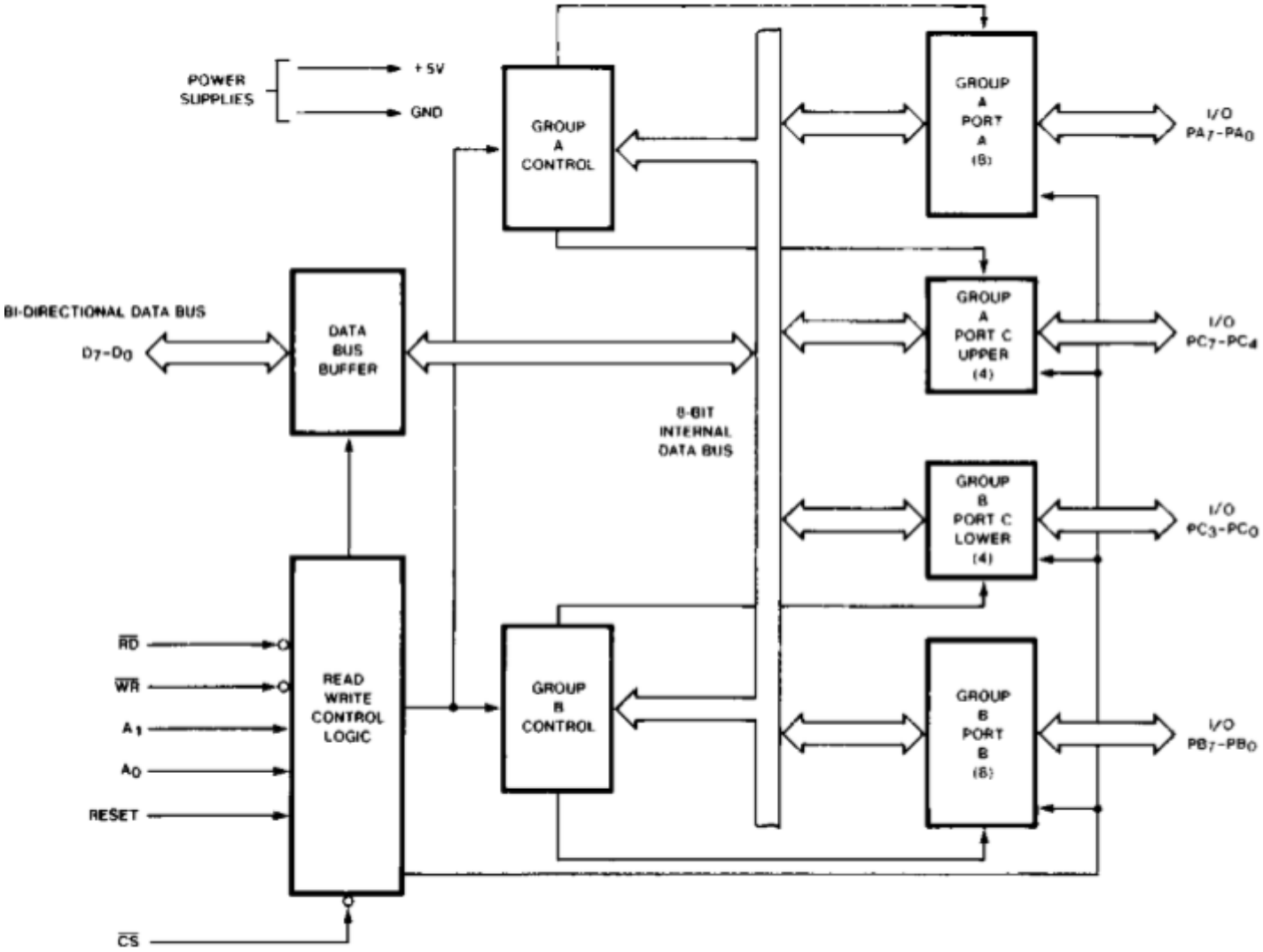
Memory mapped I/O

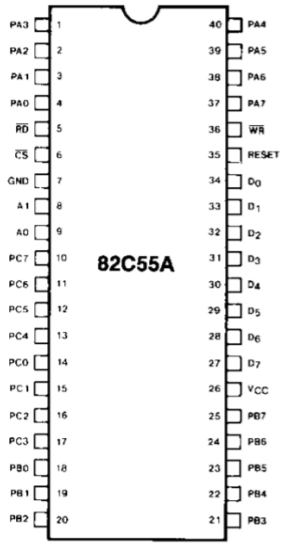


Isolated I/O or dedicated I/O



Example 8255, programmable peripheral interface





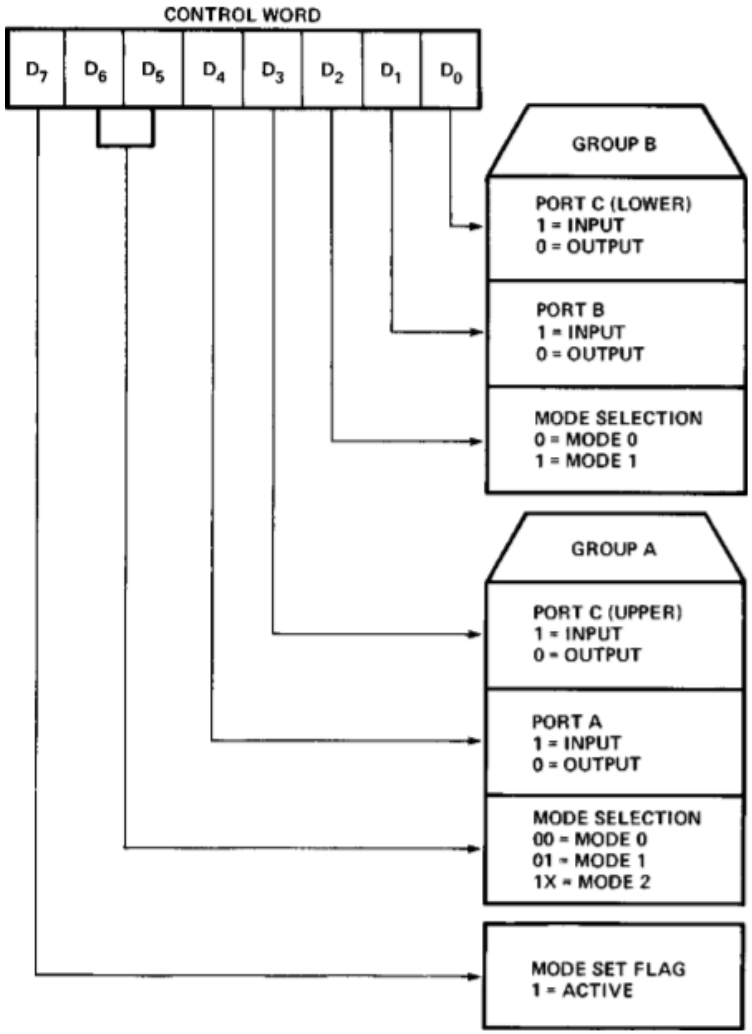
src: Intel 82C55A datasheet

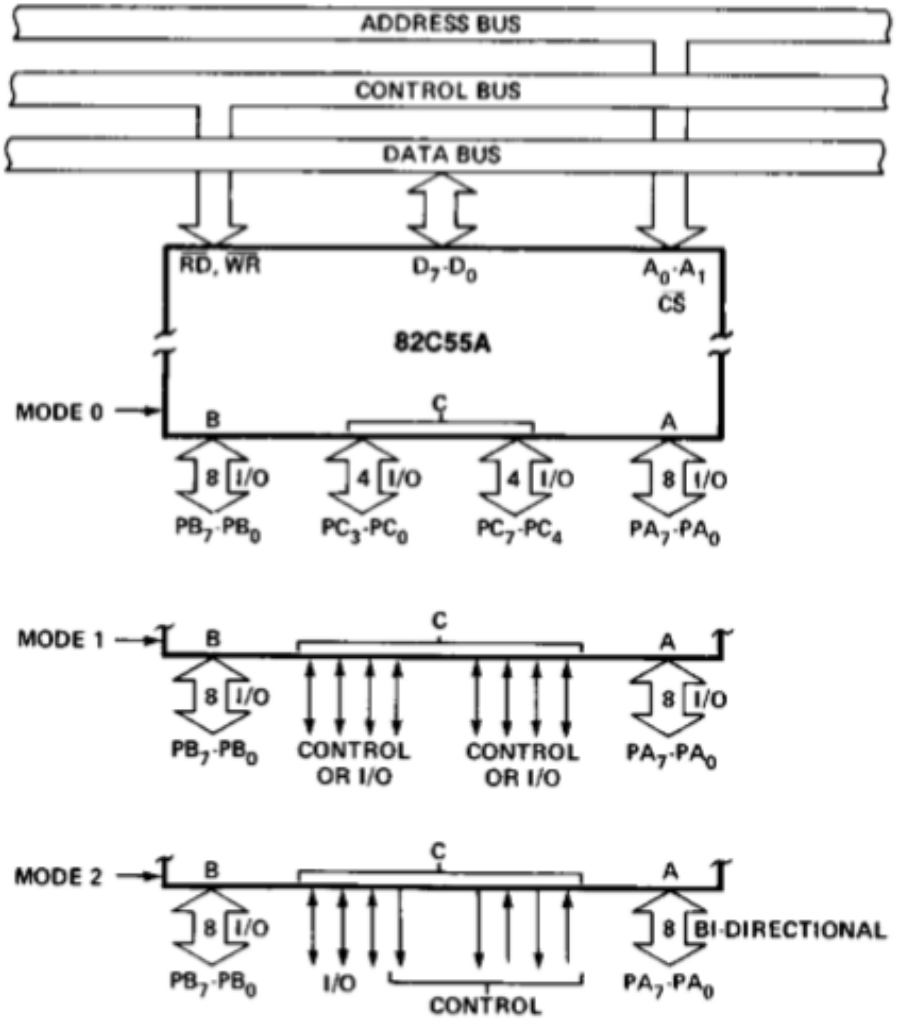
8255 address map

A_{1-0}	8–9	9–10	I	**ADDRESS:** These input signals, in conjunction $\overline{RD}$ and $\overline{WR}$, control the selection of one of the three ports or the control word registers.

A_1	A_0	$\overline{RD}$	$\overline{WR}$	$\overline{CS}$	Input Operation (Read)
0	0	0	1	0	Port A - Data Bus
0	1	0	1	0	Port B - Data Bus
1	0	0	1	0	Port C - Data Bus
1	1	0	1	0	Control Word - Data Bus
					Output Operation (Write)
0	0	1	0	0	Data Bus - Port A
0	1	1	0	0	Data Bus - Port B
1	0	1	0	0	Data Bus - Port C
1	1	1	0	0	Data Bus - Control
					Disable Function
X	X	X	X	1	Data Bus - 3 - State
X	X	1	1	0	Data Bus - 3 - State

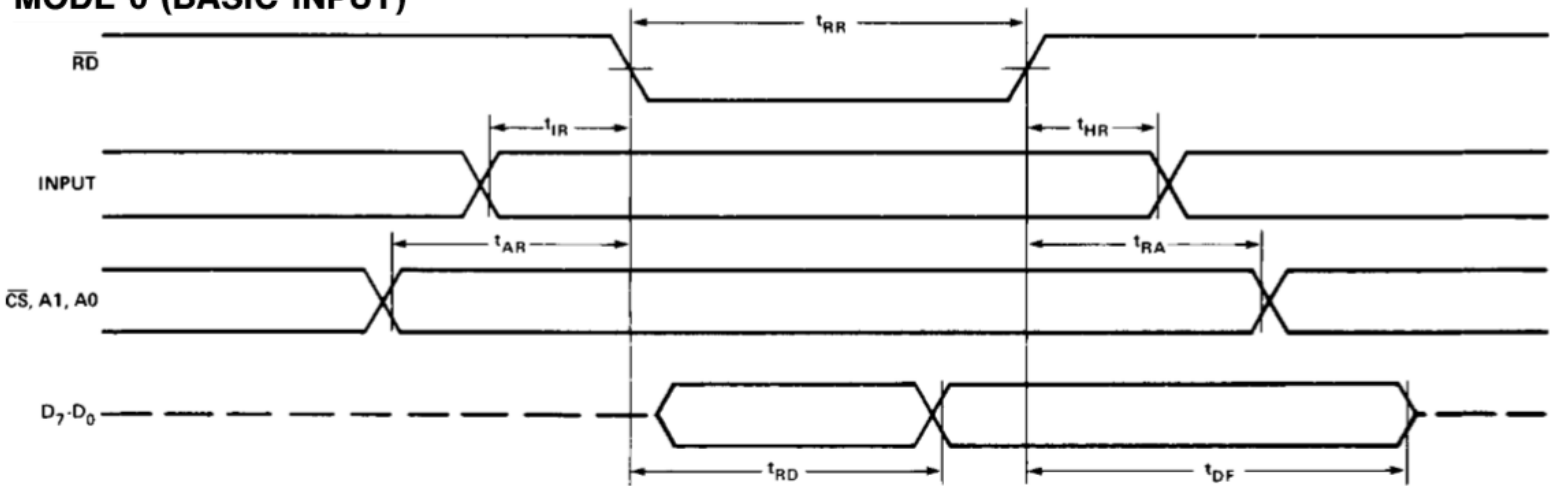
8255 modes


CONTROL WORD
D7
D6
D5
D4
D3
D2
D1
D0
GROUP B
PORT C (LOWER)
1 = INPUT
0 = OUTPUT
PORT B
1 = INPUT
0 = OUTPUT
MODE SELECTION
0 = MODE 0
1 = MODE 1
GROUP A
PORT C (UPPER)
1 = INPUT
0 = OUTPUT
PORT A
1 = INPUT
0 = OUTPUT
MODE SELECTION
00 = MODE 0
01 = MODE 1
1X = MODE 2
MODE SET FLAG
1 = ACTIVE

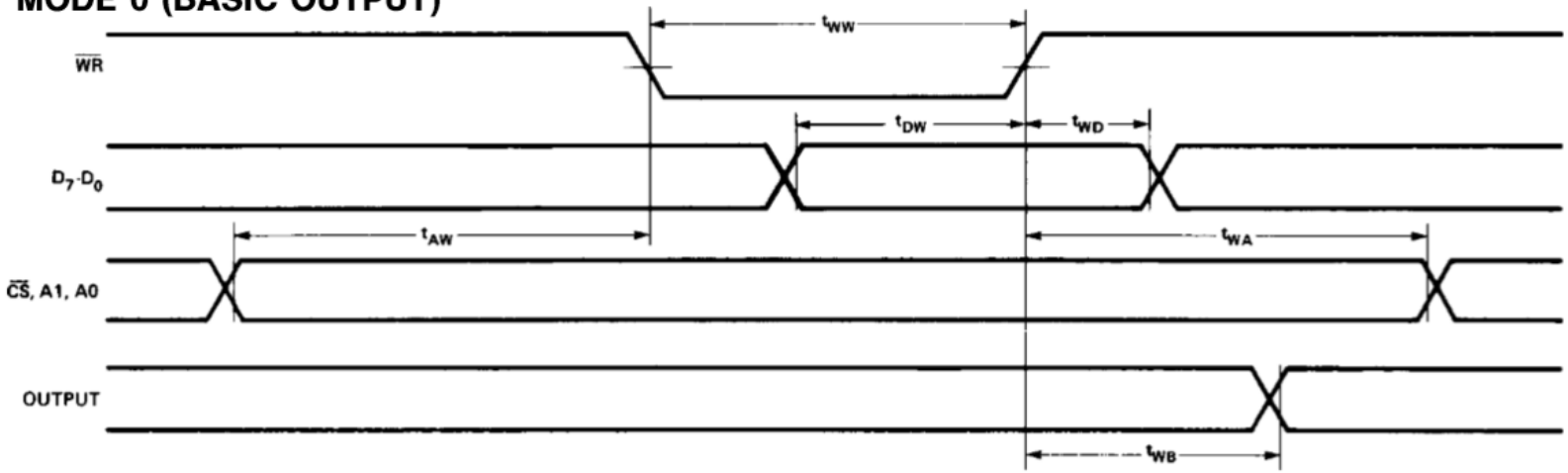

ADDRESS BUS
CONTROL BUS
DATA BUS
RD, WR
D7-D0
A0-A1
CS
82C55A
MODE 0
B
C
A
8 I/O
4 I/O
4 I/O
8 I/O
PB7-PB0
PC3-PC0
PC7-PC4
PA7-PA0
MODE 1
CONTROL OR I/O
CONTROL OR I/O
MODE 2
I/O
CONTROL
8 BI-DIRECTIONAL

8255 mode 0

MODE 0 (BASIC INPUT)



MODE 0 (BASIC OUTPUT)



8255 timing parameters (1)

READ CYCLE

Symbol	Parameter	82C55A-2		Units	Test Conditions
		Min	Max		
t_{AR}	Address Stable Before $\overline{RD}\downarrow$	0		ns	
t_{RA}	Address Hold Time After $\overline{RD}\uparrow$	0		ns	
t_{RR}	$\overline{RD}$ Pulse Width	150		ns	
t_{RD}	Data Delay from $\overline{RD}\downarrow$		120	ns	
t_{DF}	$\overline{RD}\uparrow$ to Data Floating	10	75	ns	
t_{RV}	Recovery Time between $\overline{RD}/\overline{WR}$	200		ns	

WRITE CYCLE

Symbol	Parameter	82C55A-2		Units	Test Conditions
		Min	Max		
t_{AW}	Address Stable Before $\overline{WR}\downarrow$	0		ns	
t_{WA}	Address Hold Time After $\overline{WR}\uparrow$	20		ns	Ports A & B
		20		ns	Port C
t_{WW}	$\overline{WR}$ Pulse Width	100		ns	
t_{DW}	Data Setup Time Before $\overline{WR}\uparrow$	100		ns	
t_{WD}	Data Hold Time After $\overline{WR}\uparrow$	30		ns	Ports A & B
		30		ns	Port C

8255 timing parameters (2)

OTHER TIMINGS

Symbol	Parameter	82C55A-2		Units Conditions	Test
		Min	Max		
t_{WB}	$\overline{WR}$ = 1 to Output		350	ns	
t_{IR}	Peripheral Data Before $\overline{RD}$	0		ns	
t_{HR}	Peripheral Data After $\overline{RD}$	0		ns	
t_{AK}	$\overline{ACK}$ Pulse Width	200		ns	
t_{ST}	$\overline{STB}$ Pulse Width	100		ns	
t_{PS}	Per. Data Before $\overline{STB}$ High	20		ns	
t_{PH}	Per. Data After $\overline{STB}$ High	50		ns	
t_{AD}	$\overline{ACK}$ = 0 to Output		175	ns	
t_{KD}	$\overline{ACK}$ = 1 to Output Float	20	250	ns	
t_{WOB}	$\overline{WR}$ = 1 to $\overline{OBF}$ = 0		150	ns	
t_{AOB}	$\overline{ACK}$ = 0 to $\overline{OBF}$ = 1		150	ns	
t_{SIB}	$\overline{STB}$ = 0 to IBF = 1		150	ns	
t_{RIB}	$\overline{RD}$ = 1 to IBF = 0		150	ns	
t_{RIT}	$\overline{RD}$ = 0 to INTR = 0		200	ns	
t_{SIT}	$\overline{STB}$ = 1 to INTR = 1		150	ns	
t_{AIT}	$\overline{ACK}$ = 1 to INTR = 1		150	ns	
t_{WIT}	$\overline{WR}$ = 0 to INTR = 0		200	ns	see note 1
t_{RES}	Reset Pulse Width	500		ns	see note 2

NOTE:
1. INTR ↑ may occur as early as $\overline{WR}$ ↓.
2. Pulse width of initial Reset pulse after power on must be at least 50 μSec. Subsequent Reset pulses may be 500 ns minimum. The output Ports A, B, or C may glitch low during the reset pulse but all port pins will be held at a logic "one" level after the reset pulse.

8255 mode-0 example

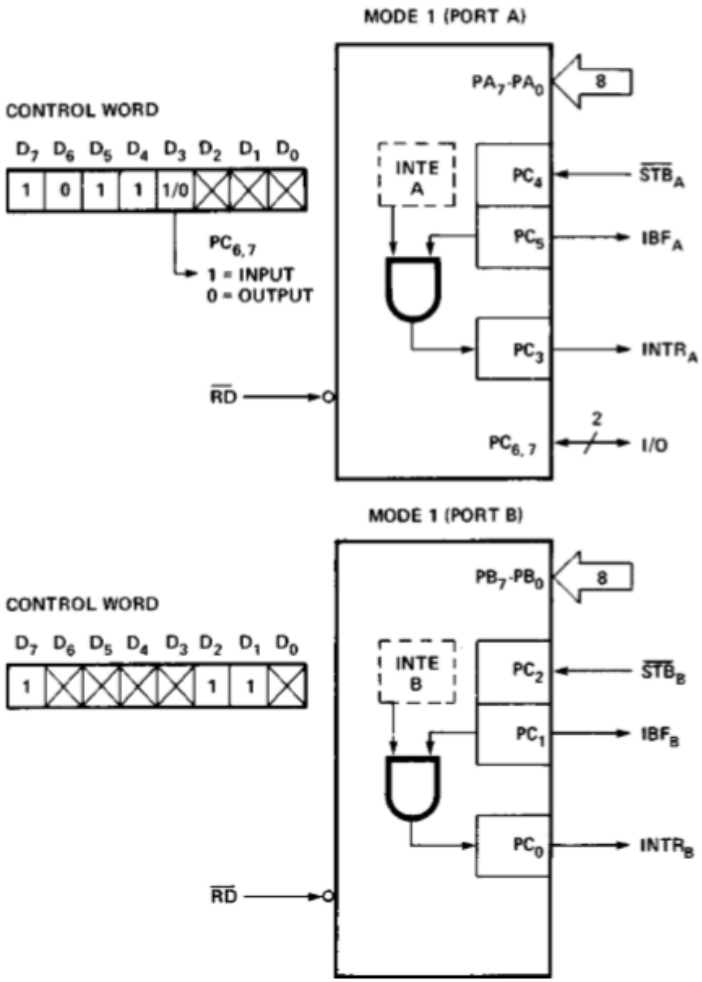
- Suppose we want to configure mode 0: port A as output, port B as input, upper port C as output, lower port C as input.
- Then, we “write” data ‘‘0b1000011’’ to address ‘‘0b11’’ (control word).
- Once initialized, we can output to port A,
  by “write” data ‘‘0b01010011’’ to address ‘‘0b00’’ (port A)
  $\Rightarrow$ output 0x53 on port A.
- Or, we can read input from port B,
  by “read” data from address ‘‘0b01’’ (port B)
  $\Rightarrow$ input data on port B.

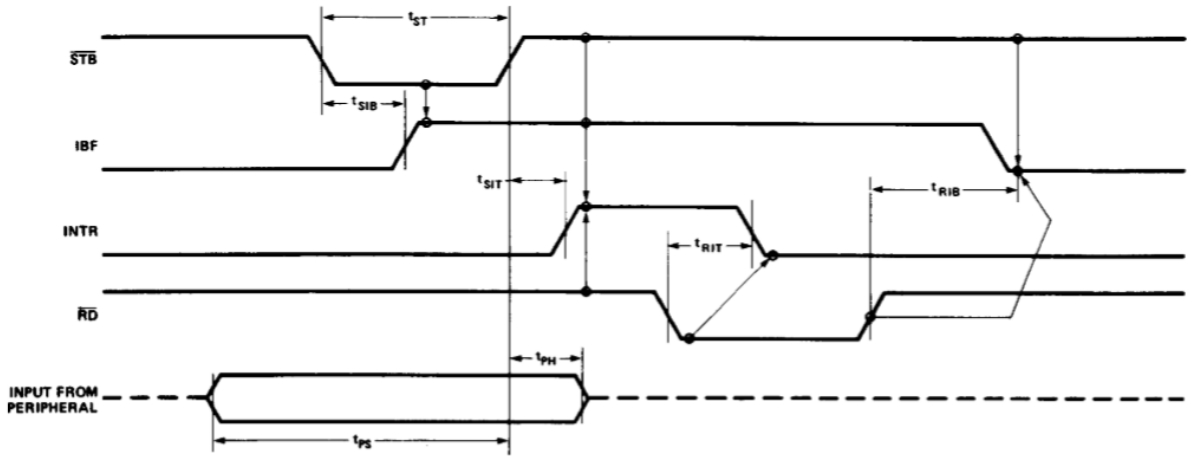
8255 questions

Read the datasheet, then answer

- Given memory-mapped scheme, how many addresses does a 8255 need?
- Suppose the I/O address space is assigned starting from 0x00500000, design an appropriate decoder to put one 8255 into the I/O address space.
- How to set Port A as output and Port B as input in mode 0?
- How can CPU send data to input port A?
- How can CPU get data from output port B?
- How fast can CPU read from 8255 input port in mode 1?
- How fast can CPU write 8255 output port in mode 1?

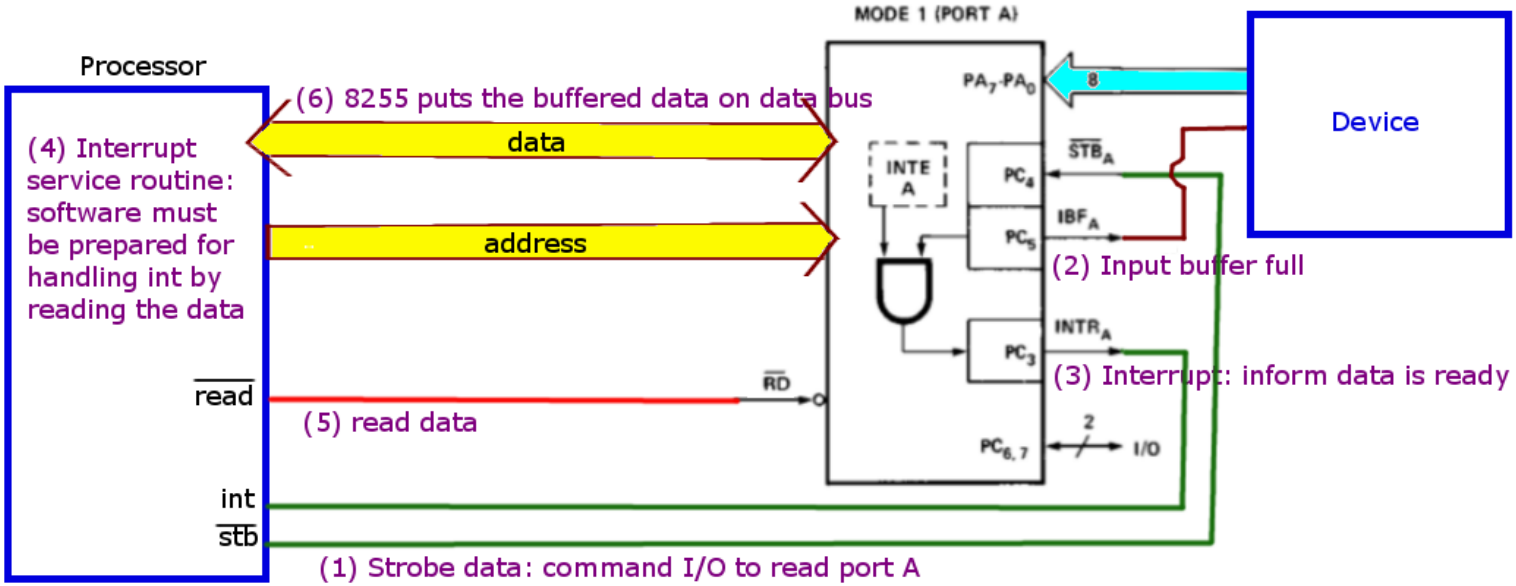
8255 mode 1 input

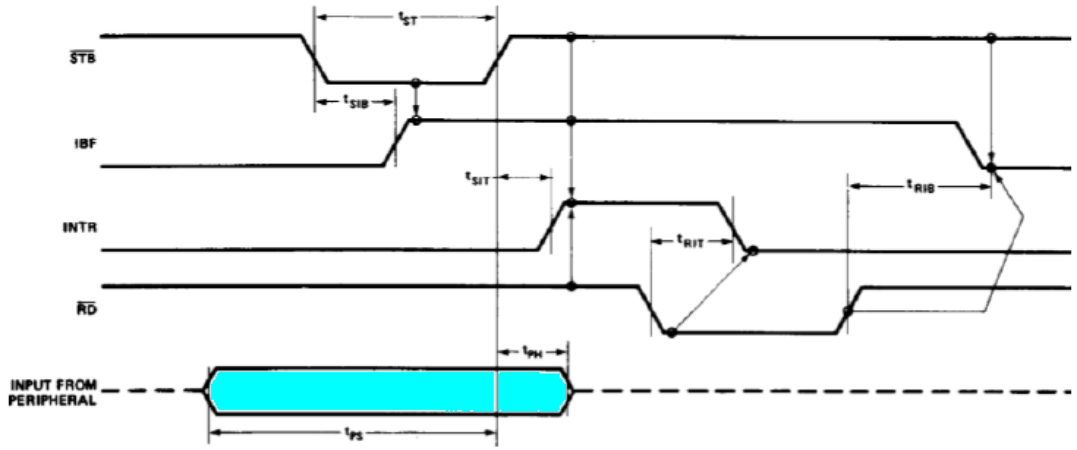


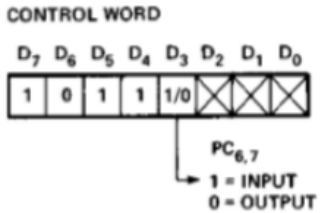


- $\overline{STB}$ to let 8255 read from its input port.
- *IBF* (input buffer full) 8255 indicates that the input port is read.
- *INTR* is to let CPU know that the data is ready.
- $\overline{RD}$ to ask 8255 to put the read input on the data bus.

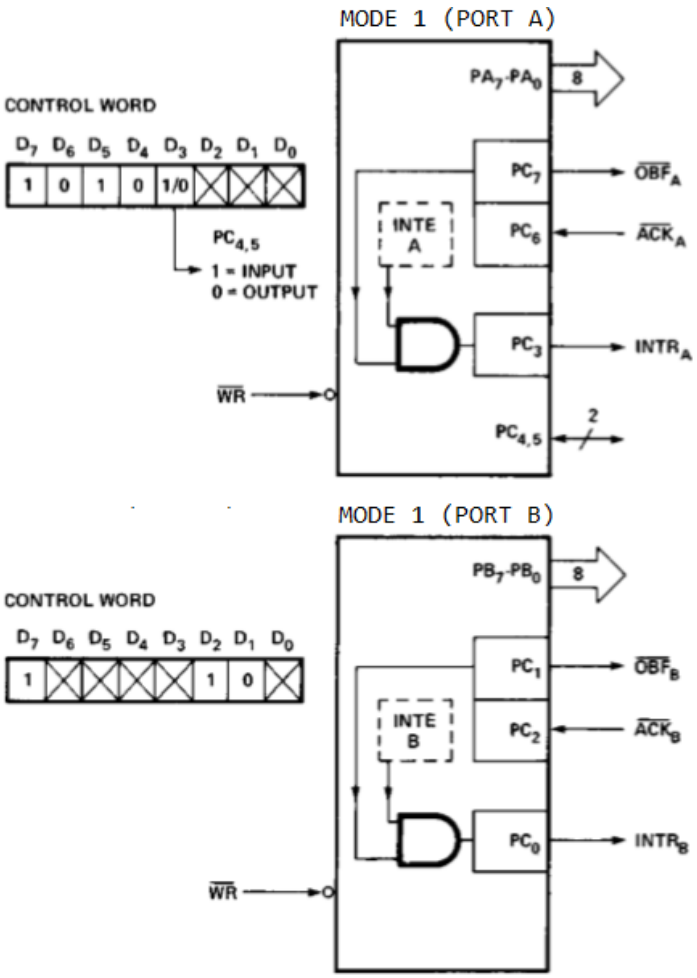
Mode-1 (8255) reads input: connection example

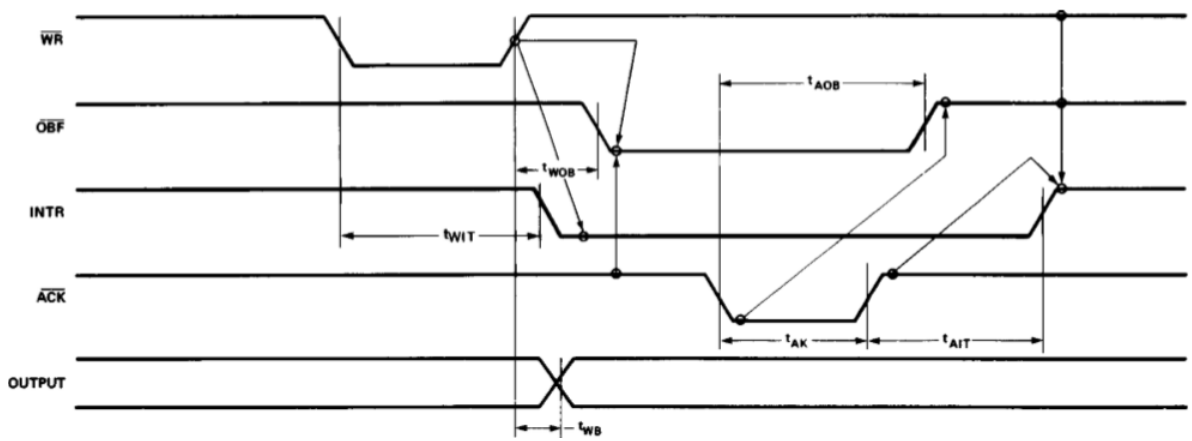






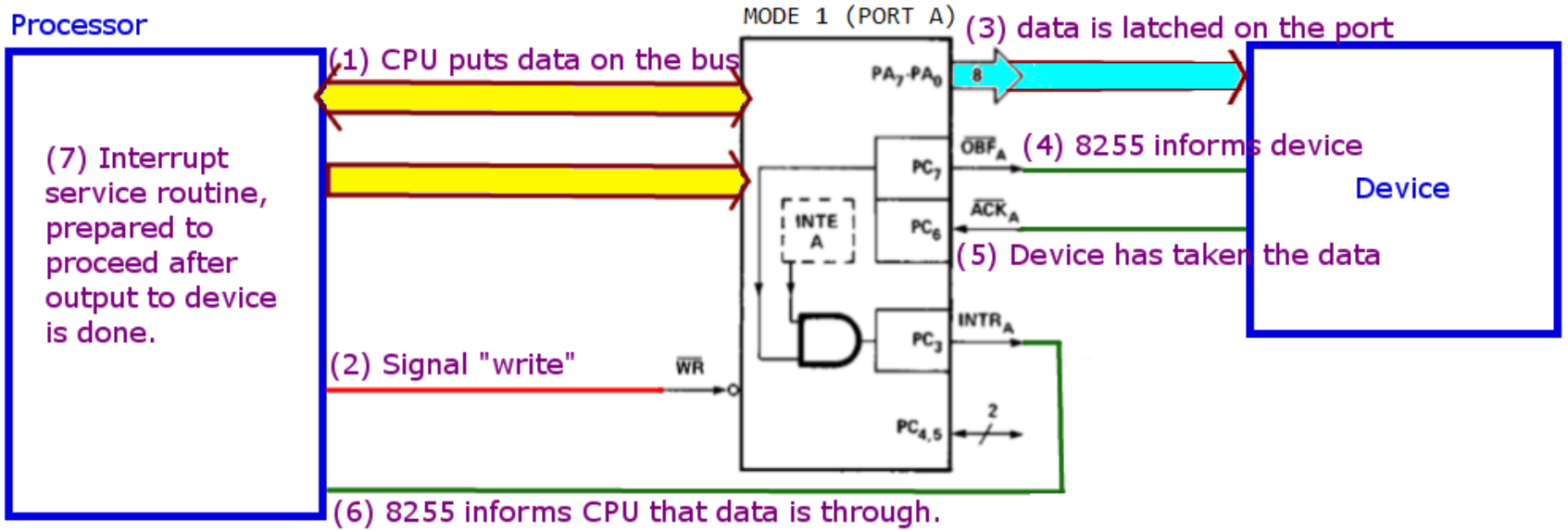
8255 mode 1 output

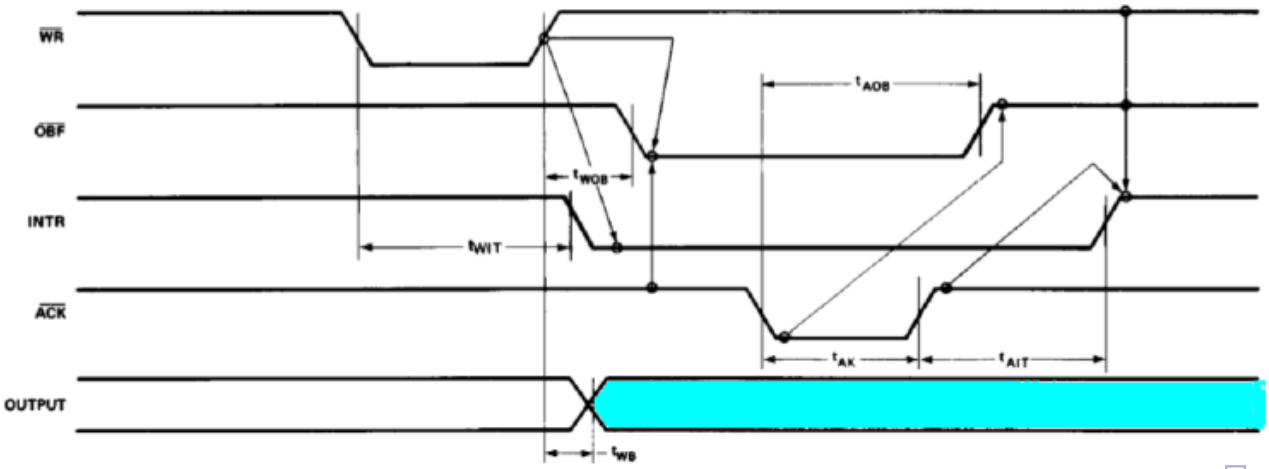


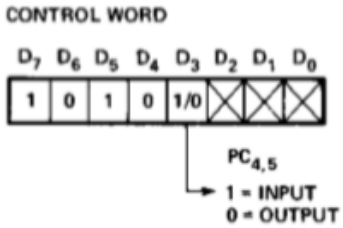


- $\overline{WR}$ to ask 8255 to pass data to the port
- $\overline{OBF}$ 8255 announces that it has gotten the data from CPU.
- $\overline{ACK}$ Peripheral informs that it has gotten the data from 8255.
- $INTR$ is to let CPU know that the data is taken.

Mode-1 (8255) writes output: connection example







8255 mode 1 questions

Read the datasheet, then answer

- How to set Port A as input and Port B as output in mode 1?
- How can CPU get data from input port A?
- How can CPU send data to output port B?
- In input mode, what may happen if peripheral data is gone in 10ns after $\overline{STB}$ rising edge?
- In input mode, what may happen if peripheral data is gone in 80ns after $\overline{STB}$ rising edge?

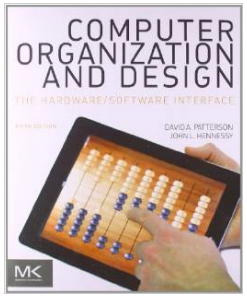
Roadmap

- Computer 101A (Done)

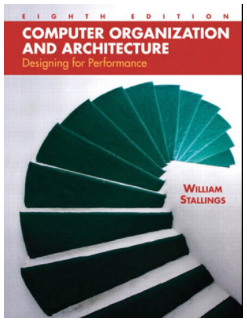
  Computer system, memory access, I/O access, and interrupt
- Computer 101B (Next)
- Instruction set
- Computer arithmetic
- Data path and control
- Pipeline
- Computer 101C

Reading

P&H's Chapter 1



Stallings's Chapter 3



Project